

Technology Stack

Date	2 July 2026
Team ID	XXXXXX
Project Name	Smaer lender
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, CSS3	Used to build a simple, responsive, and user-friendly interface for entering loan applicant details and displaying prediction results.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	Flask handles routing, processes user input, loads the trained machine learning model, and returns the prediction results to the user.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV Dataset (loan_prediction.csv)	The loan dataset is used for training the machine learning model. The trained model is stored as rdf.pkl, and the scaler is stored as scale1.pkl for prediction.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	IBM Cloud	Used to deploy the Smart Lender web application so users can access it through a web browser.

5	Version Control & CI/CD	Git & GitHub	Git is used for version control, while GitHub is used to store the project, collaborate with team members, and submit the final project repository.
	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools	Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Pickle, VS Code, Anaconda Navigator	Used for data preprocessing, visualization, machine learning model training, model serialization, and application development.
	<i>(e.g., Stripe, Twilio, Google Maps)</i>		